Introduction

The **intent-to-treat (ITT)** principle analyses participants according to their randomised assignment, irrespective of actual treatment received or adherence. It estimates the effect of *prescribing* the intervention under real-world conditions. **Per-protocol (PP)** restricts analysis to adherent participants and estimates the effect of *receiving* the intervention – a different estimand.

Prerequisites

Randomisation; estimands; missing data.

Theory

ITT preserves randomisation and tends to be conservative in superiority trials (dilution by non-adherers). PP can exaggerate efficacy or introduce selection bias because adherence is post-randomisation.

Modified ITT (mITT) excludes participants who never started or have no post-baseline data; common but can reintroduce bias if exclusion correlates with arm.

Non-inferiority trials: PP and ITT are both reported; the more conservative result (less close to margin) drives inference.

Assumptions

Randomisation is properly concealed; adherence classification is unaffected by knowledge of outcome; missingness mechanism is characterised.

R Implementation

```r
set.seed(2026)
n_per <- 100
arm <- rep(c("trt", "ctrl"), each = n_per)

# 20 % non-adherence in trt arm; 5 % in ctrl
adhered <- ifelse(arm == "trt", rbinom(n_per * 2, 1, 0.8),
                  rbinom(n_per * 2, 1, 0.95))
adhered[1:n_per] <- rbinom(n_per, 1, 0.8)
adhered[(n_per+1):(2 * n_per)] <- rbinom(n_per, 1, 0.95)

# True effect when actually received
true_effect <- ifelse(adhered == 1 & arm == "trt", 0.7, 0)
y <- rnorm(2 * n_per) + true_effect

# ITT analysis (analyse as randomised)
itt <- t.test(y ~ arm)
# Per-protocol analysis (adherers only)
pp  <- t.test(y[adhered == 1] ~ arm[adhered == 1])

rbind(ITT = c(est = diff(itt$estimate), p = itt$p.value),
      PP  = c(est = diff(pp $estimate), p = pp $p.value))
```

Output & Results

ITT effect is diluted by non-adherence; PP effect recovers the on-treatment effect but is subject to selection bias.

Interpretation

"The primary ITT analysis estimated a 0.56 point advantage for the intervention (95 % CI 0.28-0.84, $p < 0.001$); PP analysis gave 0.71 (CI 0.38-1.04). ITT is the primary inference; PP is supportive."

Practical Tips

- Pre-specify ITT as primary in the SAP; never switch post-hoc.
- Report a flow diagram (CONSORT) showing how participants were classified.
- Handle missing outcome data with multiple imputation, not naive exclusion.
- **Complier-average causal effect (CACE)** via instrumental variables estimates the effect among adherers without PP's selection bias.
- Non-inferiority trials commonly report both ITT and PP; both must meet the margin.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Diagnostic testing: Se, Sp, PPV, NPV, LR
- Kappa, ICC, Bland–Altman
- Biomarker statistics (Youden, NRI, decision curves)
- TRIPOD-AI, fairness auditing, reproducibility at scale